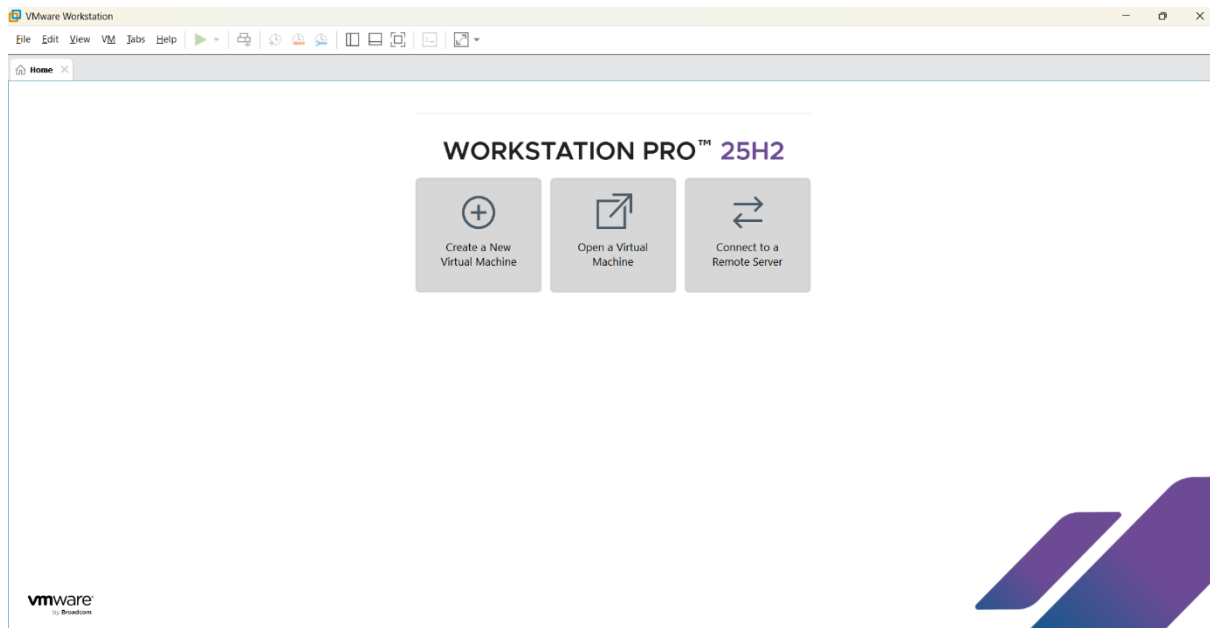


Installing Linux (Ubuntu server /Centos) on virtualbox /vmware ? basic system configuration (hostname , static ip , ssh setup)? Managing users & groups

Q.1 Install Ubuntu Server (or centos) on virtualbox or vmware , take a screenshot of your running login screen, and save it as install_success.png.

Ans-



Q.2 Change the hostname of your linux vm to myfoodiebox using the hostnamectl command , then verify the change by running the hostname command.

Ans- Step 1- Current hostname check : hostname

Step 2-change hostname : hostnamectl set-hostname myfoodiebox

Step 3- hostname

```
root@localhost:~  
[root@localhost ~]# hostname  
localhost.localdomain  
[root@localhost ~]# hostname myfoodiebox  
[root@localhost ~]# hostname  
myfoodiebox  
[root@localhost ~]#
```

**Q.3 Configure a static IP address for your VM's network interface (e.g , enp0s3 or eth0) so that it always gets the same IP on reboot.

 Hint: edit /etc/netplan/*.yaml (ubuntu) or /etc/sysconfig/network-scripts/ifcfg-eth0(centos),then restart networking.**

Ans- Step 1- nmcli device status

Step 2- vi /etc/sysconfig/network-scripts/ifcfg-ens33

Step 3- systemctl restart NetworkManager

Step 4- ip addr

```
root@localhost:~  
[root@myfoodiebox ~]# nmcli device status  
DEVICE  TYPE      STATE      CONNECTION  
ens33   ethernet  connected  ens33  
lo      loopback  connected (externally)  lo  
[root@myfoodiebox ~]# vi /etc/sysconfig/network-scripts/ifcfg-ens33  
[root@myfoodiebox ~]# systemctl restart NetworkManager  
Failed to restart NetworkManager.service: Unit NetworkManager.service not found.  
[root@myfoodiebox ~]# systemctl restart NetworkManager  
[root@myfoodiebox ~]# ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:9b:6b:db brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    inet 192.168.163.139/24 brd 192.168.163.255 scope global dynamic noprefixroute ens33  
        valid_lft 1795sec preferred_lft 1795sec  
    inet6 fe80::20c:29ff:fe9b:6bdb/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[root@myfoodiebox ~]#
```

Q.4 Enable SSH access on your linux vm and connect to it from your host machine using and SSH client (like Putty or ssh command).Show the output of whoami after login.

Ans- Step 1- install SSH server: yum install -y openssh-server

Step 2-start ssh: systemctl enable sshd

Systemctl start sshd

Step 3- check status: Systemctl status sshd

Step 4- verify login user: whoami

```
root@localhost:~  
[root@localhost ~]# yum install -y openssh-server  
CentOS Stream 9 - BaseOS                        879 kB/s | 8.9 MB    00:10  
CentOS Stream 9 - AppStream                     650 kB/s | 28 MB    00:43  
CentOS Stream 9 - Extras packages               7.6 kB/s | 21 kB    00:02  
Last metadata expiration check: 0:00:01 ago on Mon 29 Jun 2026 12:39:00 AM EDT.  
Package openssh-server-8.7p1-43.el9.x86_64 is already installed.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Upgrading:				
openssh	x86_64	9.9p1-8.el9	baseos	425 k
openssh-clients	x86_64	9.9p1-8.el9	baseos	772 k
openssh-server	x86_64	9.9p1-8.el9	baseos	546 k

```
Transaction Summary  
=====
```

Upgrade 3 Packages				
Total download size: 1.7 M				
Downloading Packages:				
(1/3): openssh-9.9p1-8.el9.x86_64.rpm	351 kB/s	425 kB	00:01	
(2/3): openssh-server-9.9p1-8.el9.x86_64.rpm	285 kB/s	546 kB	00:01	
(3/3): openssh-clients-9.9p1-8.el9.x86_64.rpm	303 kB/s	772 kB	00:02	

Total	530 kB/s	1.7 MB	00:03	
CentOS Stream 9 - BaseOS	671 kB/s	1.6 kB	00:00	

```
Importing GPG key 0x8483C65D:  
Userid : "CentOS (CentOS Official Signing Key) <security@centos.org>"  
Fingerprint: 99DB 78FA F1D7 CE22 7EB6 4882 05B5 55B3 8483 C65D
```

```
root@localhost:~  
Running transaction test  
Transaction test succeeded.  
Running transaction  
  Preparing : 1/1  
Running scriptlet: openssh-9.9p1-8.el9.x86_64 1/6  
Upgrading : openssh-9.9p1-8.el9.x86_64 1/6  
Upgrading : openssh-clients-9.9p1-8.el9.x86_64 2/6  
Running scriptlet: openssh-clients-9.9p1-8.el9.x86_64 2/6  
Running scriptlet: openssh-server-9.9p1-8.el9.x86_64 3/6  
Upgrading : openssh-server-9.9p1-8.el9.x86_64 3/6  
Running scriptlet: openssh-server-9.9p1-8.el9.x86_64 3/6  
Running scriptlet: openssh-server-8.7p1-43.el9.x86_64 4/6  
Cleanup : openssh-server-8.7p1-43.el9.x86_64 4/6  
Running scriptlet: openssh-server-8.7p1-43.el9.x86_64 4/6  
Running scriptlet: openssh-clients-8.7p1-43.el9.x86_64 5/6  
Cleanup : openssh-clients-8.7p1-43.el9.x86_64 5/6  
Cleanup : openssh-8.7p1-43.el9.x86_64 6/6  
Running scriptlet: openssh-8.7p1-43.el9.x86_64 6/6  
Verifying : openssh-9.9p1-8.el9.x86_64 1/6  
Verifying : openssh-8.7p1-43.el9.x86_64 2/6  
Verifying : openssh-clients-9.9p1-8.el9.x86_64 3/6  
Verifying : openssh-clients-8.7p1-43.el9.x86_64 4/6  
Verifying : openssh-server-9.9p1-8.el9.x86_64 5/6  
Verifying : openssh-server-8.7p1-43.el9.x86_64 6/6  
  
Upgraded:  
openssh-9.9p1-8.el9.x86_64 openssh-clients-9.9p1-8.el9.x86_64 openssh-server-9.9p1-8.el9.x86_64  
Complete!
```

```
[root@localhost ~]# systemctl enable sshd  
[root@localhost ~]# systemctl start sshd  
[root@localhost ~]# systemctl status sshd  
● sshd.service - OpenSSH server daemon  
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)  
   Active: active (running) since Mon 2026-06-29 00:39:15 EDT; 2min 19s ago  
     Docs: man:sshd(8)  
           man:sshd_config(5)  
    Main PID: 3843 (sshd)  
       Tasks: 1 (limit: 10738)  
      Memory: 1.1M  
         CPU: 57ms  
    CGroup: /system.slice/ssh.service  
            └─3843 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"  
  
Jun 29 00:39:15 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...  
Jun 29 00:39:15 localhost.localdomain sshd[3843]: Server listening on 0.0.0.0 port 22.  
Jun 29 00:39:15 localhost.localdomain sshd[3843]: Server listening on :: port 22.  
Jun 29 00:39:15 localhost.localdomain systemd[1]: Started OpenSSH server daemon.  
[root@localhost ~]# whoami  
root  
[root@localhost ~]#
```

Q.5 Create a new user named ‘Zomarouser’ and add them to a new group called ‘foodies’. Set their password and verify group membership using the groups command.

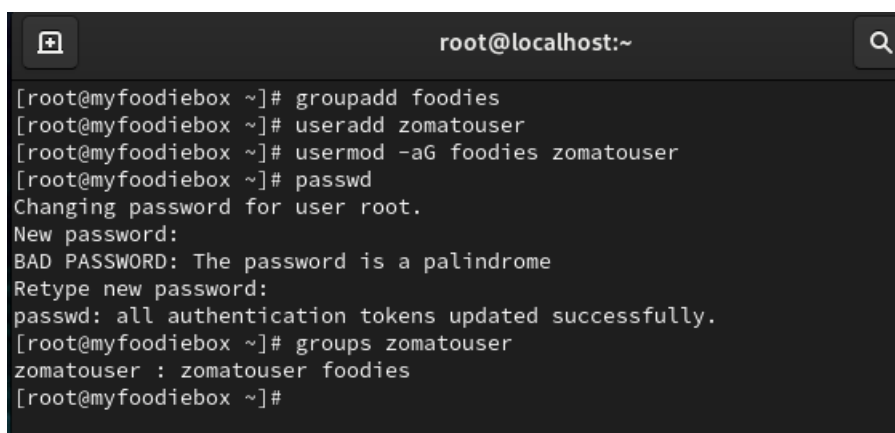
Ans- Step 1- Create group: groupadd foodies

Step 2- Create user: useradd zomatouser

Step 3-Add user to group: usermod -aG foodies zomatouser

Step 4- set passwd: passwd zomatouser

Step 5-verify group membership: groups zomatouser



```
root@localhost:~  
[root@myfoodiebox ~]# groupadd foodies  
[root@myfoodiebox ~]# useradd zomatouser  
[root@myfoodiebox ~]# usermod -aG foodies zomatouser  
[root@myfoodiebox ~]# passwd  
Changing password for user root.  
New password:  
BAD PASSWORD: The password is a palindrome  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@myfoodiebox ~]# groups zomatouser  
zomatouser : zomatouser foodies  
[root@myfoodiebox ~]#
```